Supplemental material for XYZ

1 Snow covered area and streamflow anomalies

To investigate the hydrologic ramifications of snow ablation, we obtained streamflow data from the USGS Saghen creek gage which has long term records. We found the monthly average streamflow at Saghen creek site peaked in March, two months earlier than normal (Fig. S1) and reached flows well above typical for March.

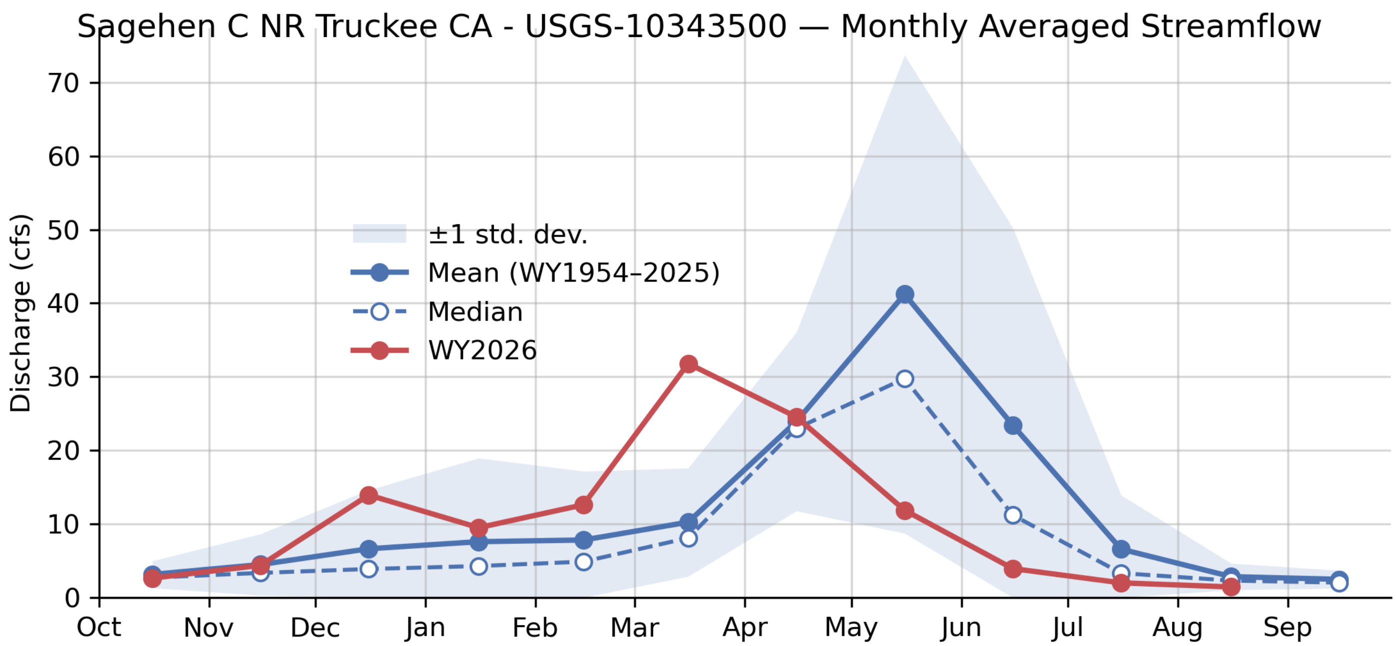

Figure S1: The monthly averaged streamflow anomalies in cubic-feet-per second (cfs) between 1954 to present for the USGS Sagehen Creek gage site. The historical mean, historical median, historical 1-standard deviation (shaded region), and WY2026 streamflow are depicted.

The synoptic pattern of snow covered area change also reveals the extent of the anomalous March. The SPIReS snow covered area (SCA) product shows the lowest snow covered area on record (Fig. S2). SCA was already very low leading up to March, but reached even lower levels by March 31. High elevation regions above 2250 m had nearly 100% SCA prior to the March event but those regions also declined substantially.

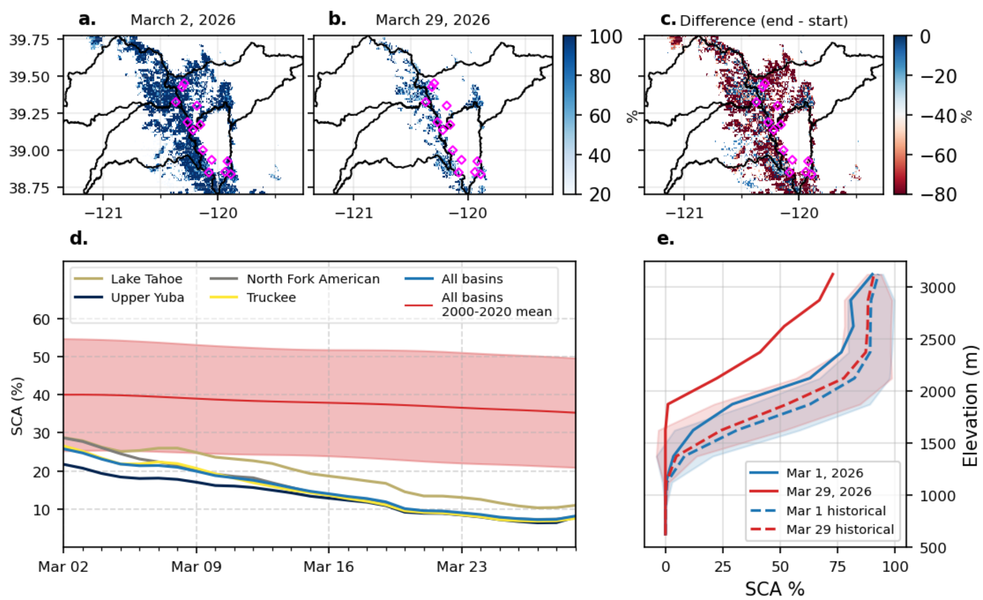

Figure S2: SPIRES SCA across the study region during the March heatwave. Watershed boundaries shown in black and SNOTEL monitoring sites indicated by magenta diamonds. **(a, b, c)** SCA (%) on March 2 and March 29, 2026, respectively, and the difference between the two dates. **(d)** Daily mean SCA (%) for each individual watershed and aggregated across all four basins. The 2000–2020 historical mean and $\pm$ 1 standard deviation are depicted. **(e)** Mean SCA as a function of elevation band (250 m bins) for March 2 (blue) and March 29 (red), shown alongside historical values for the same dates (dashed lines) with $\pm$ 1 standard deviation shading.

2 SUMMA details

This section describes assumptions implicit in SUMMA as well as provides additional details about the model forcing data preparation.

2.1 Model equations

The bulk expression for the H_s implemented in SUMMA is given by:

$$H_s = c_p \rho C_s u (T_{air} - T_{snow}), \tag{1}$$

where c_p is the specific heat capacity of dry air (constant), u is the horizontal wind speed, ρ is the dry air density, and C_s is the bulk transport coefficient. In this work, we use the SUMMA formulation for C_s, which models the stability correction as a function of the bulk Richardson number.

We use the "Loisinv" stability correction function implemented in the SUMMA model and assume that $z_h = z_0$

$$C_h = \frac{k^2}{\ln(z/z_0)^2} \cdot f(\mathrm{Ri}) \tag{2}$$

Where Ri is the bulk Richardson number, defined as

$$\mathrm{Ri} = \frac{9.81(T_{air} - T_{snow})}{0.5(T_{air} + T_{snow})u^2}. \tag{3}$$

and f(Ri) is a function with range (0,1] that declines as a function of increasing Ri. An equivalent formulation for latent heat flux follows Eq. 1. The stability correction is reproduced and used in Fig. 11 of the main text.

2.2 Snow albedo

$$\mathrm{age1} = \exp\big[-\beta,(Tfreeze - T_{surf})\big], \qquad \mathrm{age}_2 = \mathrm{age}_1^{;10}$$

$$\mathrm{decayFactor} = \frac{\Delta t,(\mathrm{age}_1 + \mathrm{age}_2 + \mathrm{age}_3)}{\mathrm{albedoDecayRate}}$$

2.3 Forcing data

We use bias corrected ERA5 data as forcing for SUMMA with several key steps implemented. ERA5 data was obtained from the Copernicus data store. TODO: WHAT GRID CELL

$SW\downarrow$ was compared between ERA and CSSL CNR4 radiometers for water year 2026. We developed a correction factor to account for potential shadowing by canopy adjacent to the site, using solar zenith angle (SZA) and solar azimuth angle (AZM) as predictors. These were computed from the python py-ephem package TODO: CITE. The correction grid is depicted in Fig. S3. It shows the multiplicative correction between 0 and 1 for each SZA and AZM angle. The correction reduced $SW\downarrow$ by TODO: XYZ for the month of March 2026.

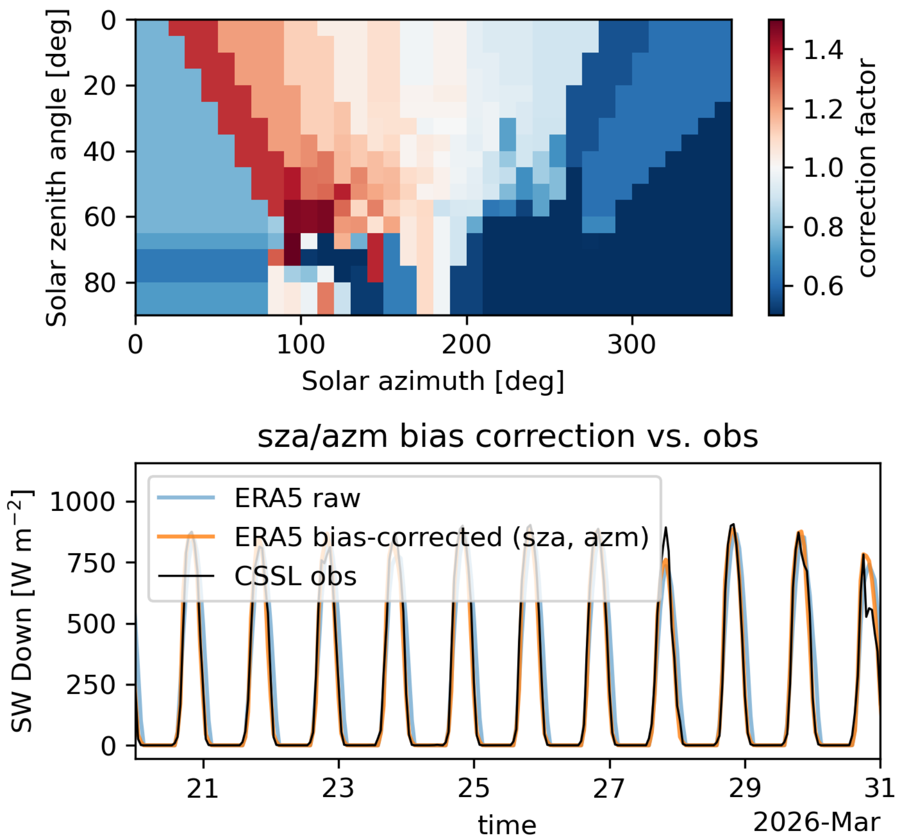

Figure S3: $SW\downarrow$ multiplicative correction factors for the ERA5 radiation data. The top plot shows the correction grid. The bottom plot shows the observed, uncorrected, and corrected data.

2.4 Model validation

2.4.1 Canopy site

We validate the SUMMA model in several ways. First, we evaluated the canopy radiation parameterization against the CSSL sub-canopy Kipp and Zonene SPLite 2 radiometer located under the canopy. We find good correlation for March of 2026 for the UEB 2-stream parameterzation and an LAI value of 1.5 (Fig S5), using the aforementioned $SW \downarrow$ forcing as the top boundary condition to the model.

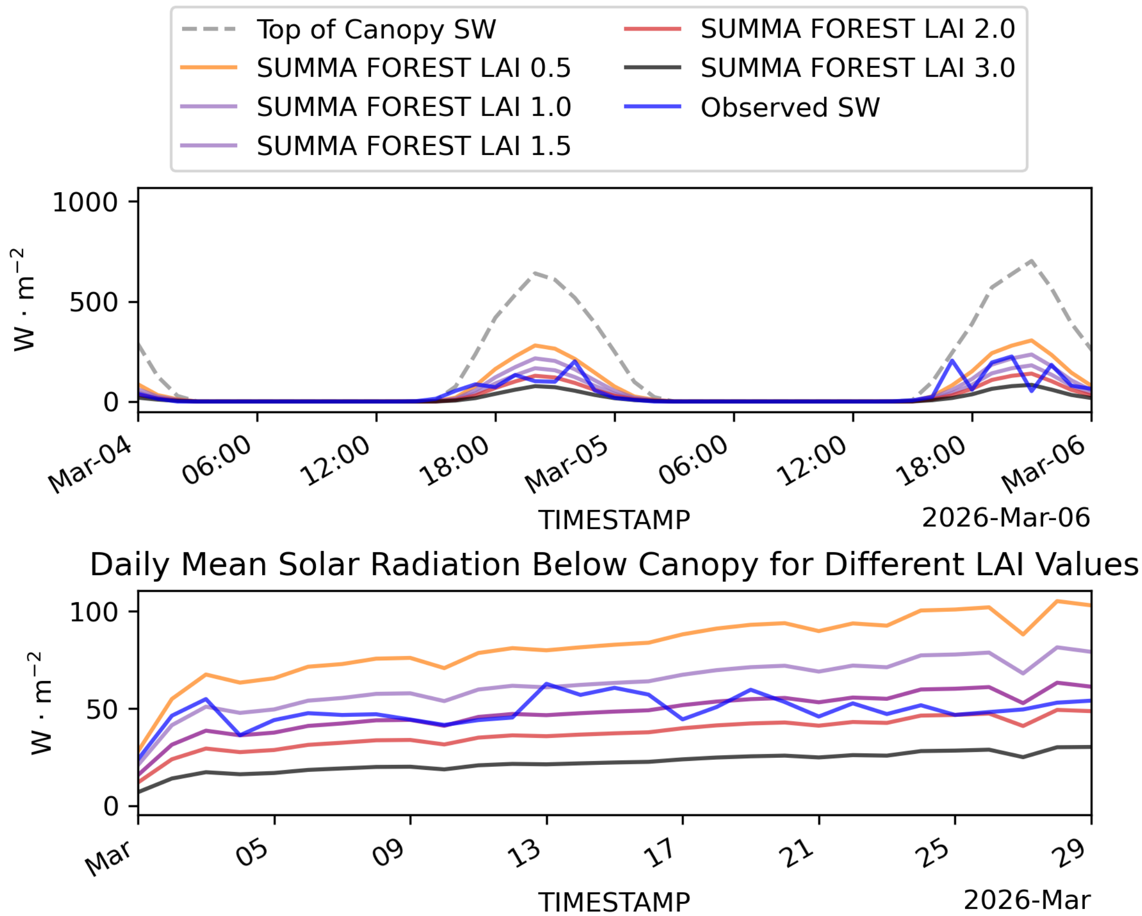

Figure S4: A comparison of sub-canopy downwelling radiation for different LAI values using the UEB-2stream shortwave radiation parameterization. The top plot depicts a hourly SW↓ two day period in early March. The bottom plot depicts hourly SW↓ during March. Observed radiation by the Kipp and Zonen radiometer is depicted

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2.4.2 Fluxes

CSSL hosts an experimental eddy-covariance instrument on a movable arm above the snowpack (see Fig. 1 in the main text). Eddy covariance fluxes were processed using the Licor EddyPro v7.0.9 software with despiking, double coordinate rotation, and flux quality assignment. TODO: X% of the H_s and H_l fluxes are retained for analysis during the month of March. A qualitative comparison between the SUMMA and eddy-covariance fluxes of sensible and latent heat show reasonable agreement, though there were many low-quality flux measurements during the month of March. Key periods of sublimation during the first and latter portions of the the month show reasonable agreement.

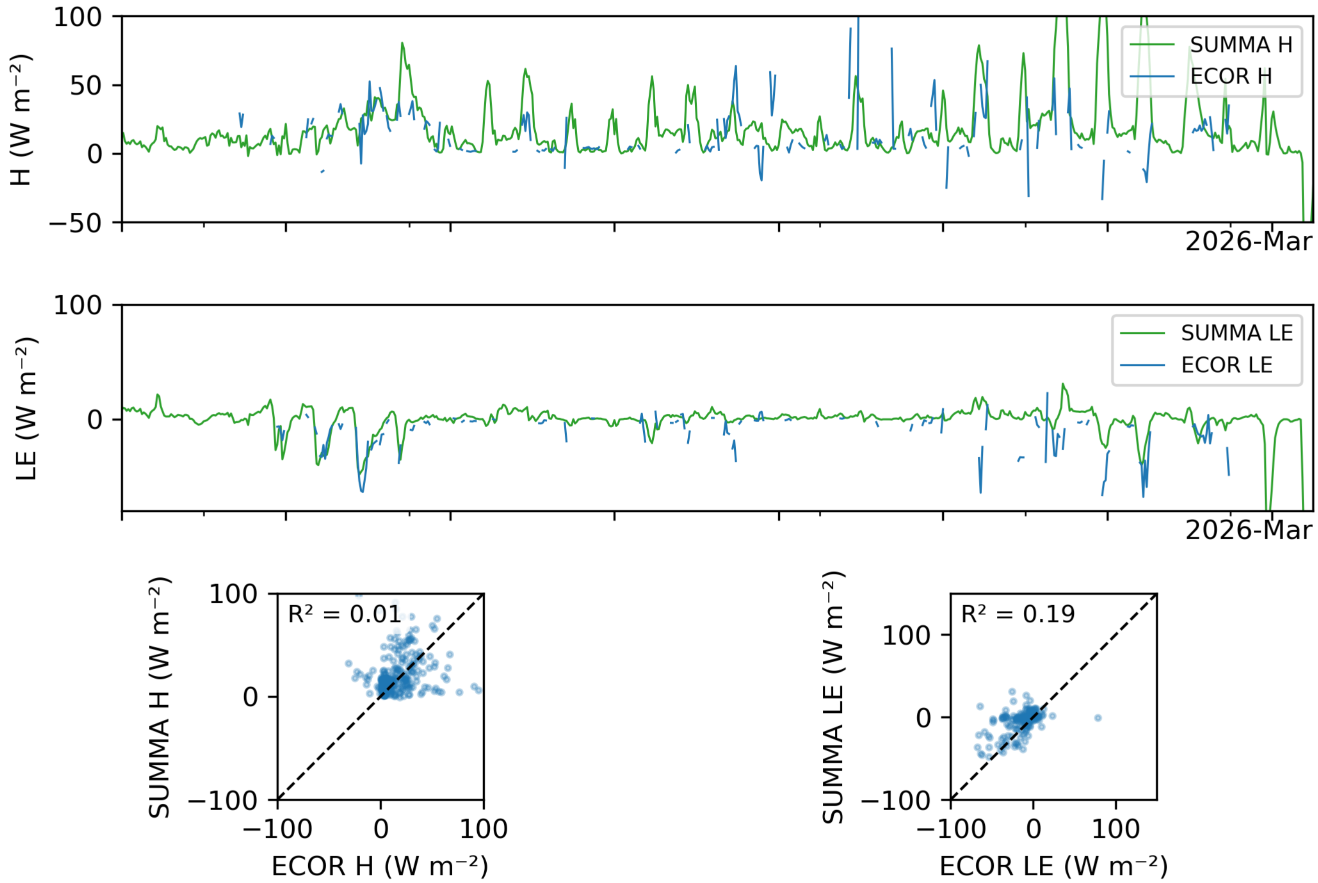

Figure S5: A comparison of SUMMA and eddy-covariance (ECOR) measured fluxes at CSSL for sensible heat (H) and latent heat (LE). The top shows a timeseries for the month of March 2026. The bottom two plots show 1:1 plots between the ECOR and SUMMA estimated fluxes.

3 Initial condition experiments

As we have shown, anomolous conditions also preceded the March 2026 ablation event. To interrogate the role of antecedent conditions, we performed initial condition experiments by altering the amount of ice, liquid water, and cold-content. We did three experiments. One is the natural starting condition (liq=1). Then we halved and zeroed out the amount of liquid water and increased the amount ice to keep SWE constant. For internal consistency in SUMMA the layer temperature reduced in the zero-liquid water case. The outcomes show that there is minimal sensitivity to the timing in the rate of SWE change; the all-ice case completely ablated about one-day later than the standard case.

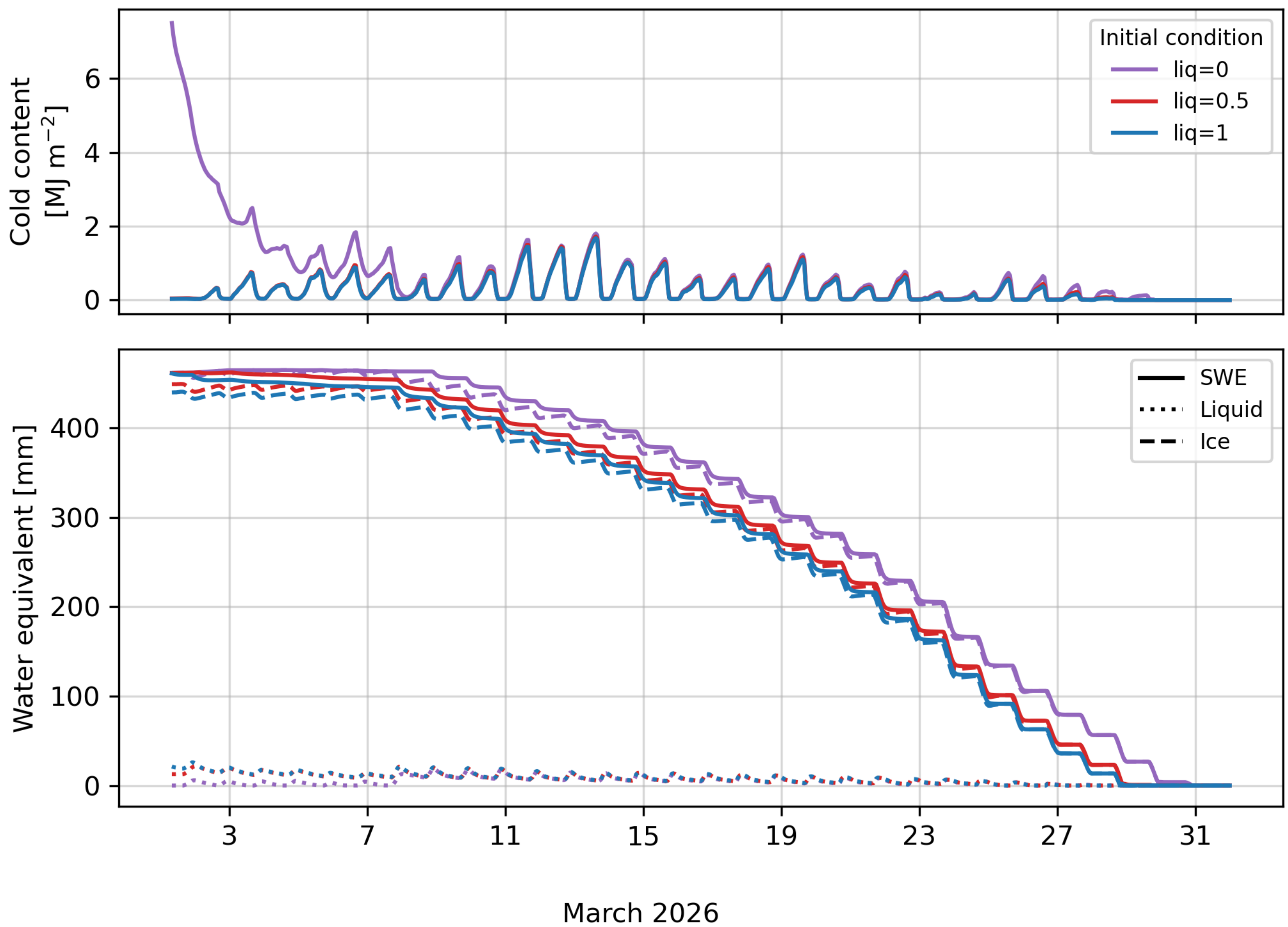

Figure S6: Initial condition tests. (a) a schematic of the snow mass balance composed of ice and liquid water, (b) the initial and final SWE and snow depth, and (c) mass components of the change in SWE for the different experiments.